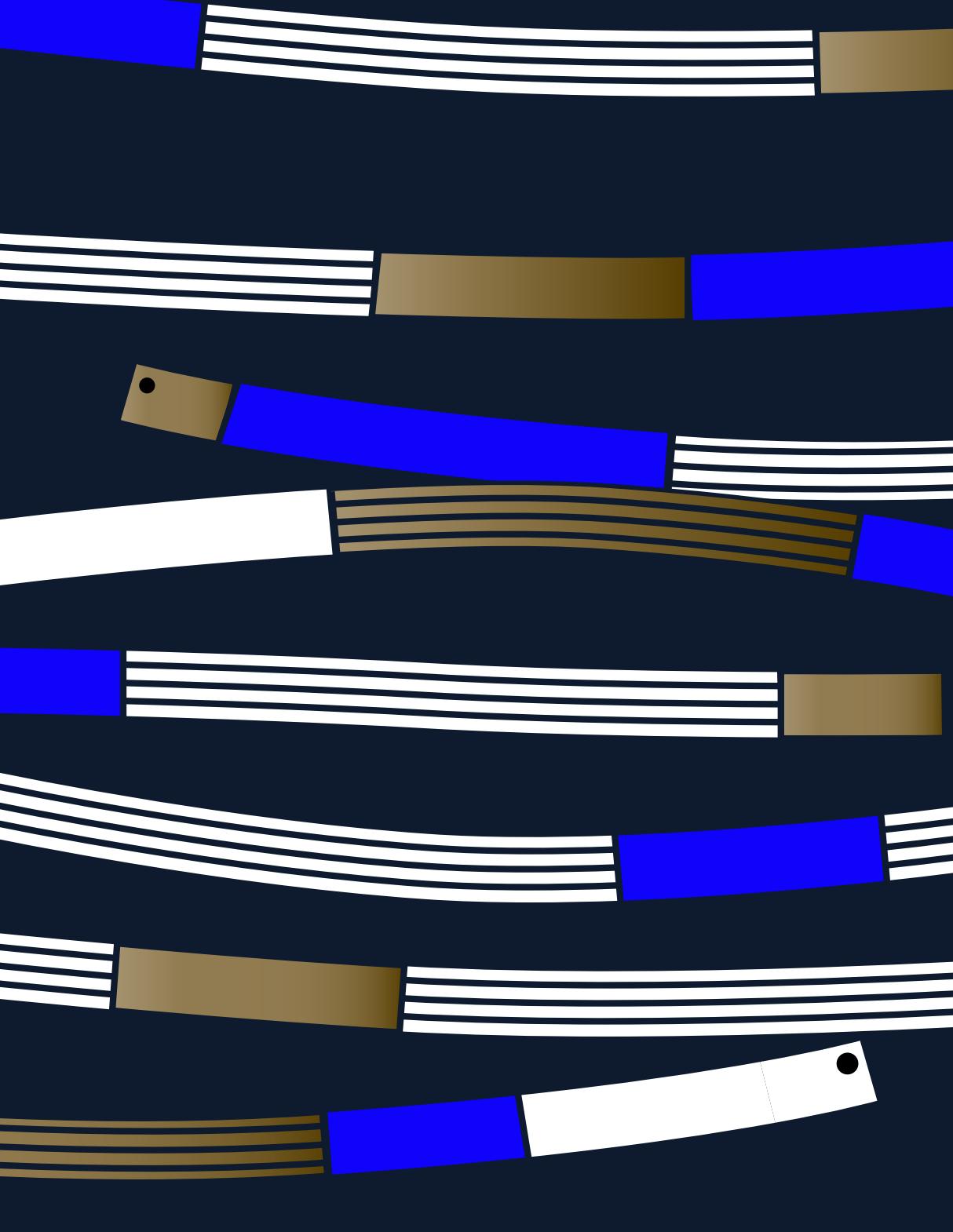




RMC Homework-5

1. (a) $L \frac{di}{dt} + Ri + \frac{1}{C} \int i dt = u, \left(\frac{1}{C} \int i dt = y \right)$

$$Ls I(s) + R I(s) + \frac{1}{C} \cdot \frac{I(s)}{s} = U(s) \quad \frac{1}{C} \frac{I(s)}{s} = Y(s)$$

$$\rightarrow I(s) = Cs \cdot Y(s)$$

$$G(s) = \frac{Y(s)}{U(s)} = \frac{1}{Lcs^2 + Rcs + 1}$$

$$LC \ddot{y} + Rc \ddot{y} + y = u \quad \text{or} \quad \ddot{y} + \frac{R}{L} \dot{y} + \frac{1}{LC} y = \frac{u}{LC}$$

$$y = x_1$$

$$\dot{y} = \dot{x}_1 = x_2$$

$$\ddot{y} = \dot{x}_2 = -\frac{1}{LC} x_1 - \frac{R}{L} x_2 + \frac{1}{LC} u$$

$$\dot{x} = \begin{bmatrix} 0 & 1 \\ -1/LC & -R/L \end{bmatrix} x + \begin{bmatrix} 0 \\ 1/LC \end{bmatrix} u$$

$$y = \begin{bmatrix} 1 & 0 \end{bmatrix} x$$

(b) $\frac{1}{C_1} \int (i_1 - i_2) dt + R_1 i_1 = u$

$$\frac{1}{C_1} \int (i_2 - i_1) dt + R_2 i_2 + \underbrace{\frac{1}{C_2} \int i_2 dt}_{\rightarrow y} = 0$$

Applying Laplace transformation,

$$\frac{I_1(s) - I_2(s)}{sC_1} + R_1 I_1(s) = U(s)$$

$$\frac{I_2(s) - I_1(s)}{sC_1} + R_2 I_2(s) + \underbrace{\frac{I_2(s)}{sC_2}}_{\rightarrow y(s)} = 0 \quad (1)$$

$$I_1(s) \left[\frac{1}{sC_1} + R_1 \right] + \frac{I_2(s)}{sC_1} = U(s)$$

$$I_1(s) = \frac{U(s) - \frac{I_2(s)}{sC_1}}{\frac{1}{sC_1} + R_1} \quad (2)$$

Sub (2) in (1) & $G(s) = \frac{Y(s)}{U(s)}$

1b) continued...

$$G(s) = \frac{1}{\underbrace{R_1 C_1 R_2 C_2 s^2}_A + \underbrace{(R_1 C_1 + R_2 C_2 + R_1 C_2)s + 1}_B}$$

$$A\ddot{y} + B\dot{y} + y = u$$

$$\text{let } y = x_1, \dot{y} = \dot{x}_1 = x_2,$$

$$\ddot{y} = \dot{x}_2 = -\frac{x_1}{A} - \frac{x_2 B}{A} + \frac{u}{A}$$

$$\dot{x} = \begin{bmatrix} 0 & 1 \\ -1/A & -B/A \end{bmatrix} x + \begin{bmatrix} 0 \\ 1/A \end{bmatrix} u$$

Replacing c_1 by L & let $c_2 = C$.

By applying loop law;

$$i_1 R_1 + L \left(\frac{di_1}{dt} - \frac{di_2}{dt} \right) = u$$

$$i_2 R_2 + \frac{1}{C} \int i_2 dt + L \left(\frac{di_2}{dt} - \frac{di_1}{dt} \right) = 0.$$

$$\text{so, } \frac{1}{C} \int i_2 dt = y.$$

By Applying Laplace,

$$R_1 I_1(s) + Ls [I_1(s) - I_2(s)] = U(s)$$

$$R_2 I_2(s) + \frac{1}{C} \frac{I_2(s)}{s} + Ls [I_2(s) - I_1(s)] = 0.$$

$$\text{Then, } G(s) = y(s)/U(s) = \frac{Ls}{\underbrace{CL(R_1 + R_2)s^2}_A + \underbrace{(R_1 R_2 C + L)s + R_1}_B}$$

$$y(s) = \frac{Ls U(s)}{As^2 + Bs + R_1} \rightarrow w(s)$$

$$\frac{w(s)}{U(s)} = \frac{1}{As^2 + Bs + R_1}$$

$$y(s) = Ls w(s)$$

$$A\ddot{w} + B\dot{w} + R_1 w = u$$

$$\text{and } y = L\dot{w}$$

$$w = x_1$$

$$\dot{w} = \dot{x}_1 = x_2$$

$$\ddot{w} = \dot{x}_2 = -\frac{R_1 x_1}{A} - \frac{B x_2}{A} + \frac{u}{A}$$

$$\dot{x} = \begin{bmatrix} 0 & 1 \\ -R_1/A & -B/A \end{bmatrix} x + \begin{bmatrix} 0 \\ 1/A \end{bmatrix} u$$

$$y = \begin{bmatrix} 0 & L \end{bmatrix} x.$$

$$2) \quad m_1 \ddot{x}_1 = -K_1 x_1 - K_2 (x_1 - x_2) - b(\dot{x}_1 - \dot{x}_2) + u$$

$$m_2 \ddot{x}_2 = -K_3 x_2 - K_2 (x_2 - x_1) - b(\dot{x}_2 - \dot{x}_1)$$

Taking Laplace transformations, we obtain $x_1(s)$ and $x_2(s)$ in terms of $U(s)$

Then it's easy to get,

$$m_1 s^2 x_1(s) = -K_1 x_1(s) - K_2 [x_1(s) - x_2(s)] - bs[x_1(s) - x_2(s)] \quad \text{--- (1)}$$

$$m_2 s^2 x_2(s) = -K_3 x_2(s) - K_2 [x_2(s) - x_1(s)] - bs[x_2(s) - x_1(s)] \quad \text{--- (2)}$$

Substituting x_2 from (2) and (1) we get,

$$\frac{x_1(s)}{U(s)} = \frac{m_2 s^2 + bs + K_2 + K_3}{(m_1 s^2 + bs + K_1 + K_2)(m_2 s^2 + bs + K_2 + K_3) - (bs + K_2)^2} \quad \text{Substitute } x_1(s) \text{ from (1) in (2),}$$

$$\frac{x_2(s)}{U(s)} = \frac{bs + K_2}{(m_1 s^2 + bs + K_1 + K_2)(m_2 s^2 + bs + K_2 + K_3) - (bs + K_2)^2}$$

$$3) \quad (a) \quad \dot{x} = Ax + Bu \quad A = \begin{bmatrix} -5 & -1 \\ 3 & -1 \end{bmatrix}, \quad B = \begin{bmatrix} 2 \\ 5 \end{bmatrix}, \quad C = [1 \ 2]$$

$$y = Cx$$

$$G(s) = C(sI - A)^{-1}B = [1 \ 2] \begin{bmatrix} s+5 & 1 \\ -3 & s+1 \end{bmatrix}^{-1} \begin{bmatrix} 2 \\ 5 \end{bmatrix} = \frac{12s + 59}{s^2 + 6s + 8}$$

(b) For finding step input,

$$x(t) = e^{At} x(0) + \int_0^t e^{A(t-\tau)} B u(\tau) d\tau; \quad \text{where } x(0) = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$e^{At} = L^{-1} [sI - A]^{-1} = L^{-1} \begin{bmatrix} s+5 & 1 \\ -3 & s+1 \end{bmatrix}^{-1}$$

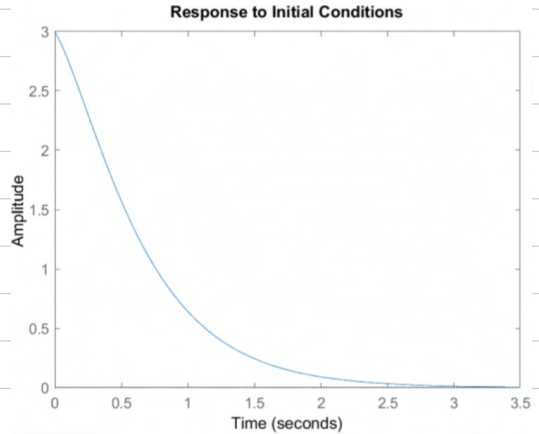
$$= \begin{bmatrix} -\frac{1}{2} e^{-2t} + \frac{3}{2} e^{-4t} & -\frac{1}{2} e^{-2t} + \frac{1}{2} e^{-4t} \\ \frac{3}{2} e^{-2t} - \frac{3}{2} e^{-4t} & \frac{3}{2} e^{-2t} - \frac{1}{2} e^{-4t} \end{bmatrix}$$

$$x(t) = e^{At} x(0) = \begin{bmatrix} e^{-2t} + 2e^{-4t} \\ 3e^{-2t} - 2e^{-4t} \end{bmatrix}$$

$$y(t) = Cx(t) = 5e^{-2t} - 2e^{-4t}$$

3b) continued...

```
>> A = [-5 -1; 3 -1];
B = [2; 5];
C = [1 2];
D = 0;
sys = ss(A, B, C, D);
x0 = [1; 1];
>> initial(sys, x0)
```



$$(c) \quad A(\delta) = \begin{bmatrix} -5+\delta & -1 \\ 3 & -1 \end{bmatrix}$$

$$\det [\lambda I - A(\delta)] = \lambda^2 + (6-\delta)\lambda + 8-\delta$$

$$\delta < 6 \text{ \& } \delta < 8$$

$\therefore$ for transfer function to be stable $\delta < 8$

$$4) \quad G(s) = \frac{25}{s^2 + 4s + 25} \quad \zeta = 0.4, \omega_n = 5, K=1$$

$$= \frac{K\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

$$C(s) = P(s) = A(s) = \frac{q(s)}{s}$$

$$G(s) = \frac{C(s)}{P(s)} \rightarrow C(s) = R(s) \cdot G(s) = \frac{\omega_n^2}{s(s^2 + 2\zeta\omega_n s + \omega_n^2)}$$

$$C(s) = \frac{1}{s} - \frac{s + 2\zeta\omega_n}{s^2 + 2\zeta\omega_n s + \omega_n^2} = \frac{1}{s} - \frac{s + 3\omega_n}{(s + \zeta\omega_n)^2 + \omega_d^2} - \frac{3\omega_n}{(s + \zeta\omega_n)^2 + \omega_d^2}$$

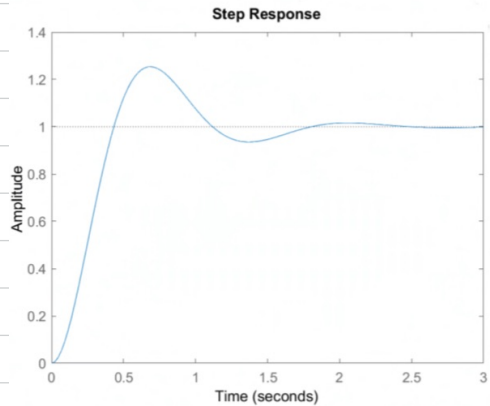
4) continued ...

$$c(t) = 1 - e^{-\zeta \omega_d t} \cos \omega_d t - \frac{\zeta}{\omega_d} e^{-\zeta \omega_d t} \sin \omega_d t, \quad \omega_d = \sqrt{1 - \zeta^2}$$

$$= 1 - e^{-\zeta \omega_d t} \left(\cos \omega_d t - \frac{\zeta}{\sqrt{1 - \zeta^2}} \sin \omega_d t \right); \text{ upon substitution,}$$

$$c(t) = 1 - e^{-2t} \left[\cos(0.91t) - \frac{0.4}{0.91} \sin(0.91t) \right]$$

```
num = [0 0 25];
den = [1 4 25];
step(num, den)
```



5) (a) propagation equations:

$$f_{i-1,i} - f_{i,i+1} + m_i g - m_i v_{c1} = 0$$

$$N_{i-1,i} - N_{i,i+1} + r_{i,c1} \cdot f_{i,i+1} - r_{i-1,c1} \cdot f_{i-1,i} - I_i \dot{w}_i - w_i \cdot (I_i \dot{w}_i) = 0$$

for link 1,

$$f_{0,1} - f_{1,2} + m_1 g - m_1 v_{c1} = 0 \quad \text{--- (1)}$$

$$N_{0,1} - N_{1,2} + r_{1,c1} \cdot f_{1,2} - r_{0,c1} \cdot f_{0,1} - I_1 \dot{w}_1 = 0 \quad \text{--- (2)}$$

for link 2,

$$f_{1,2} + m_2 g - m_2 v_{c2} = 0 \quad \text{--- (3)}$$

$$N_{1,2} + r_{1,c2} \cdot f_{1,2} - I_2 \dot{w}_2 = 0 \quad \text{--- (4)}$$

$$N_{1,2} = T_{1,2}$$

5 (a) continued...

substituting $f_{1,2}$ from (3) in (4)

$$\tau_2 - r_{1,c_2} \cdot m_2 \dot{v}_{c_2} + r_{1,c_2} \cdot m_2 g - I_2 \dot{\omega}_2 = 0$$

similarly,

$$\tau_1 - \tau_2 - r_{0,c_1} \cdot m_1 \dot{v}_{c_1} - r_{0,1} \cdot m_2 \dot{v}_{c_2} + r_{0,c_1} \cdot m_1 g + r_{0,1} \cdot m_2 g - I_1 \dot{\omega}_1 = 0$$

Now,

$$\omega_1 = \dot{\theta}_1 \quad \text{and} \quad \omega_2 = \dot{\theta}_1 + \dot{\theta}_2$$

If we assume COM of links at the ends of the links for simplicity,

$$I_1 = I_2 = 0 \quad \text{and} \quad l_{c_1} = l_1 ; \quad l_{c_2} = l_2$$

$$\text{and} \quad v_{c_1} = \begin{bmatrix} -l_{c_1} \dot{\theta}_1 \sin \theta_1 \\ l_{c_1} \dot{\theta}_1 \cos \theta_1 \end{bmatrix} ; \quad v_{c_2} = \begin{bmatrix} (l_1 \sin \theta_1 + l_{c_2} \sin \theta_2) \dot{\theta}_1 - l_{c_2} \sin \theta_2 \dot{\theta}_2 \\ (l_1 \cos \theta_1 + l_{c_2} \cos \theta_2) \dot{\theta}_1 + l_{c_2} \cos \theta_2 \dot{\theta}_2 \end{bmatrix}$$

Then,

$$\tau_1 = H_{11} \ddot{\theta}_1 + H_{12} \ddot{\theta}_2 - h \dot{\theta}_2^2 - 2h \dot{\theta}_1 \dot{\theta}_2 + g_1$$

$$\tau_2 = H_{22} \ddot{\theta}_2 + H_{12} \ddot{\theta}_1 + h \dot{\theta}_1^2 + g_2$$

where,

$$H_{11} = m_1 l_{c_1}^2 + I_1 + m_2 [l_1^2 + l_{c_2}^2 + 2l_1 l_{c_2} c_2] + I_2$$

$$H_{22} = m_2 l_{c_2}^2 + I_2$$

$$H_{12} = m_2 l_1 l_{c_2} c_2 + m_2 l_{c_2}^2 + I_2 ; \quad h = m_2 l_1 l_{c_2} s_2$$

$$g_1 = m_1 l_{c_1} g \cos \theta_1 + m_2 g [l_{c_2} \cos(\theta_{12}) + l_1 \cos \theta_1] ; \quad g_2 = m_2 l_{c_2} g \cos \theta_{12}$$

general eqn of $\tau = M(q) \ddot{q} + V(q, \dot{q}) + G(q)$

$$\begin{bmatrix} \tau_1 \\ \tau_2 \end{bmatrix} = \begin{bmatrix} (m_1 + m_2) l_1^2 + m_2 l_2^2 + 2m_2 l_1 l_2 c_2 & m_2 l_2^2 + m_2 l_1 l_2 c_2 \\ m_2 l_2^2 + m_2 l_1 l_2 c_2 & m_2 l_2^2 \end{bmatrix} \begin{bmatrix} \ddot{\theta}_1 \\ \ddot{\theta}_2 \end{bmatrix} + \begin{bmatrix} -m_2 l_1 l_2 (2\dot{\theta}_1 \dot{\theta}_2 + \dot{\theta}_2^2) s_2 \\ m_2 l_1 l_2 \dot{\theta}_1^2 s_2 \end{bmatrix} + \begin{bmatrix} (m_1 + m_2) g l_1 \cos \theta_1 + m_2 g l_2 \cos \theta_{12} \\ m_2 g l_2 \cos \theta_{12} \end{bmatrix}$$

$$(b) \ddot{\theta} = \underbrace{M^{-1}(\theta) \cdot [\tau - v(\theta, \dot{\theta}) - g(\theta)]}_u$$

$$\text{Then, } \ddot{\theta}_1 = u_1; \ddot{\theta}_2 = u_2$$

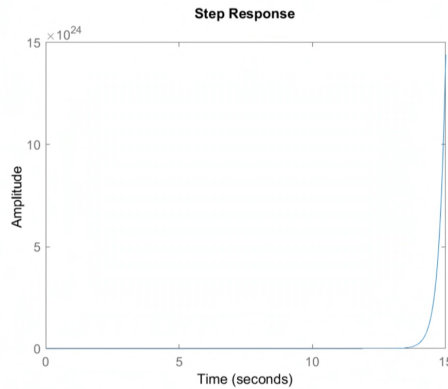
$$\theta_1 = x_1, \dot{\theta}_1 = x_2; \theta_2 = x_3; \dot{\theta}_2 = x_4$$

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \\ \dot{x}_4 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} + \begin{bmatrix} 0 & 0 \\ 1 & 0 \\ 0 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}$$

we can re-order the state variables to get
we can write it as,

$$\dot{x} = \begin{bmatrix} 0 & I \\ 0 & 0 \end{bmatrix} x + \begin{bmatrix} 0 \\ I \end{bmatrix} u \quad (u \text{ has non linear terms.})$$

(c) Assuming, $l_1 = l_2 = m_1 = m_2 = 1$



$${}^0_1 T = \begin{bmatrix} c_1 & -s_1 & 0 & 0 \\ s_1 & c_1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

α_{i-1}	α_{i-1}	d_i	Q_i
0	0	0	θ_1
θ_0	l_1	0	θ_2
0	l_2	0	0

6) continued...

$${}^1_2T = \begin{bmatrix} c_2 & -s_2 & 0 & l_1 \\ 0 & 0 & -1 & 0 \\ s_2 & c_2 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$${}^2_3T = \begin{bmatrix} 1 & 0 & 0 & l_2 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$${}^1P_2 = L_1 \hat{x}_1 ; \quad {}^2P_3 = L_2 \hat{x}_2$$

$${}^1P_1 = L_1 \hat{x}_1 ; \quad {}^2P_2 = L_2 \hat{x}_2 ; \quad {}^1I_1 = {}^2I_2 = 0 ; \quad {}^1V_0 = g \hat{z}_0$$

$$w_0 = \dot{w}_0 = 0 ; \quad F_2 = 0$$

velocity propagation:

$${}^1w_1 = {}^0R \dot{w}_0 + \dot{\theta}_1 \hat{z}_1 = \dot{\theta}_1 \hat{z}_1 = [0 \ 0 \ \dot{\theta}_1]^T$$

$${}^1\dot{w}_1 = {}^0R \ddot{w}_0 + {}^0R \dot{w}_0 \dot{\theta}_1 \hat{z}_1 + \dot{\theta}_1 \hat{z}_1 = \dot{\theta}_1 \hat{z}_1 \Rightarrow [0 \ 0 \ \ddot{\theta}_1]^T$$

$${}^1\dot{v}_1 = {}^0R [{}^0w_0 \cdot {}^0p_1 + {}^0w_0 ({}^0w_0 \cdot {}^0p_2) + \ddot{v}_0]$$

$${}^1\dot{v}_1 = \begin{bmatrix} c_1 & s_1 & 0 \\ -s_1 & c_1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ g \end{bmatrix} = [0 \ 0 \ g]^T$$

$${}^1\dot{v}_{c1} = {}^1\dot{w}_1 \cdot {}^1p_{c1} + {}^1w_1 ({}^1w_1 \cdot {}^1p_{c1}) + {}^1\dot{v}_1$$

$${}^1\dot{v}_{c1} = L_1 \ddot{\theta}_1 \hat{y}_1 + \dot{\theta}_1 \hat{z}_1 \cdot L_1 \dot{\theta}_1 \hat{y}_1 + g \hat{z}_1 = \begin{bmatrix} -L_1 \dot{\theta}_1^2 \\ L_1 \dot{\theta}_1 \\ g \end{bmatrix}$$

for link 2.

$${}^2w_2 = {}^2_1R {}^1w_1 + \dot{\theta}_2 \hat{z}_2 = \begin{bmatrix} c_2 & 0 & s_2 \\ -s_2 & 0 & c_2 \\ 0 & -1 & 0 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ \dot{\theta}_1 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ \dot{\theta}_2 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ \ddot{\theta}_2 \end{bmatrix} =$$

$${}^2w_2 = [s_2 \dot{\theta}_1 \quad c_2 \dot{\theta}_1 \quad \dot{\theta}_2]^T$$

$${}^2\dot{w}_2 = {}^2_1R {}^1\dot{w}_1 + {}^2_1R {}^1w_1 \cdot \dot{\theta}_2 \hat{z}_2 + \ddot{\theta}_2 \hat{z}_2$$

6) continued...

$${}^2\ddot{w}_2 = \begin{bmatrix} s_2 \ddot{\theta}_1 \\ c_2 \ddot{\theta}_1 \\ 0 \end{bmatrix} + \begin{bmatrix} c_2 \dot{\theta}_1 \dot{\theta}_2 \\ -s_2 \dot{\theta}_1 \dot{\theta}_2 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ \ddot{\theta}_2 \end{bmatrix} = \begin{bmatrix} s_2 \ddot{\theta}_1 + c_2 \dot{\theta}_1 \dot{\theta}_2 \\ c_2 \ddot{\theta}_1 + s_2 \dot{\theta}_1 \dot{\theta}_2 \\ \ddot{\theta}_2 \end{bmatrix}$$

$$2\dot{V}_2 = {}^2P({}^1\dot{w}_1, {}^1P_2 + {}^1w_1({}^1w_1, {}^1P_2) + \dot{V}_1)$$

$$2\dot{V}_2 = \begin{bmatrix} -L_1 c_2 \dot{\theta}_1^2 + s_2 g & L_1 s_2 \dot{\theta}_1^2 + c_2 g & -L_1 \ddot{\theta}_1 \end{bmatrix}^T$$

$$2\dot{V}_2 = {}^2w_2 \cdot {}^2P_{C_2} + {}^2w_2 ({}^2w_2 \cdot {}^2P_{C_2}) + {}^2\dot{V}_2$$

$${}^2\dot{V}_{C_2} = \begin{bmatrix} -(L_1 c_2 + L_2 c_2^2) \dot{\theta}_1^2 - L_2 \dot{\theta}_2^2 + s_2 g \\ (L_1 s_2 + L_2 s_2 c_2) \dot{\theta}_1^2 + L_2 \dot{\theta}_2^2 + c_2 g \\ 2L_2 s_2 \dot{\theta}_1 \dot{\theta}_2 - L_1 \ddot{\theta}_1 - L_2 c_2 \ddot{\theta}_1 \end{bmatrix}$$

$$\text{Then, } {}^1F_1 = M_1 {}^1\dot{V}_{C_1} = \begin{bmatrix} -M_1 L_1 \dot{\theta}_1^2 \\ M_1 L_1 \ddot{\theta}_1 \\ M_1 g \end{bmatrix}$$

$${}^1N_1 = {}^c I_1 {}^1\dot{w}_1 + {}^1w_1 {}^c I_1 {}^1w_1 = \begin{bmatrix} 0 & 0 & 0 \end{bmatrix}^T$$

link-2

$${}^2F_2 = M_2 {}^2\dot{V}_{C_2} = \begin{bmatrix} -m_2(l_1 + l_2 c_2) c_2 \dot{\theta}_1^2 - m_2 l_2 \dot{\theta}_2^2 + m_2 s_2 g \\ m_2(l_1 + l_2 c_2) s_2 \dot{\theta}_1^2 + m_2 l_2 \dot{\theta}_2^2 + m_2 s_2 g \\ 2m_2 l_2 s_2 \dot{\theta}_1 \dot{\theta}_2 - m_2 l_1 \ddot{\theta}_1 - m_2 l_2 c_2 \ddot{\theta}_1 \end{bmatrix}$$

$${}^2N_2 = {}^c I_2 {}^2\dot{w}_2 + {}^2w_2 ({}^c I_2 {}^2w_2) = \begin{bmatrix} 0 & 0 & 0 \end{bmatrix}^T$$

backward force :

$$\text{link 2)} \quad {}^2n_2 = {}^2N_2 + {}^2P {}^3n_3 + {}^2P_{C_2} {}^2F_2 + {}^2P_3 {}^2P {}^3F_3$$

$${}^2n_2 = L_2 \hat{x}_2 \cdot {}^2F_2 = \begin{bmatrix} m_2 l_1 l_2 \ddot{\theta}_1 - 2m_2 l_2^2 s_2 \dot{\theta}_1 \dot{\theta}_2 + m_2 l_2^2 c_2 \ddot{\theta}_1 \\ m_2 l_2 (l_1 + l_2 c_2) s_2 \dot{\theta}_1^2 - m_2 g l_2 c_2 \\ + m_2 l_2^2 \ddot{\theta}_2 \end{bmatrix}$$

link ①

$${}^1n_1 = {}^1N_1 + {}^1P_2 {}^2n_2 + {}^1P_{C1} {}^1F_1 + {}^1P_{22} {}^2F_2$$

$${}^1n_1 = \begin{bmatrix} (\\ (\\ (m_1 l_1^2 + m_2 l_1^2 + m_2 l_2^2 c_2 + 2m_2 l_1 l_2 c_2) \ddot{\theta}_1 - 2(l_1 + l_2 c_2) m_2 l_2 s_2 \dot{\theta}_1 \dot{\theta}_2 \end{bmatrix}$$

joint torque eqⁿ:

$$\tau_1 = {}^1n_1 {}^1\hat{z}_1$$

$$\tau_1 = [m_1 l_1^2 + m_2 (l_1 + l_2 c_2)^2] \ddot{\theta}_1 - 2(l_1 + l_2 c_2) m_2 l_2 s_2 \dot{\theta}_1 \dot{\theta}_2 + v_1 \dot{\theta}_1$$

$$\tau_2 = m_2 l_2^2 \ddot{\theta}_2 + (l_1 + l_2 c_2) m_2 l_2 s_2 \dot{\theta}_1^2 + m_2 g l_2 c_2 + v_2 \dot{\theta}_2$$

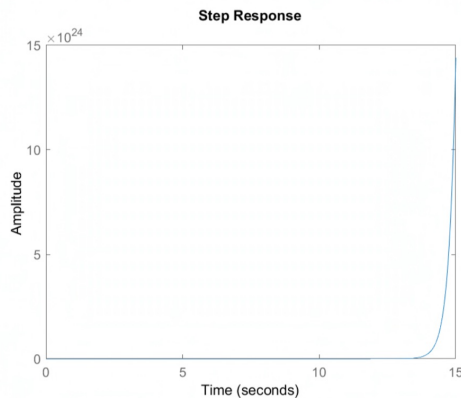
$$\tau = M(\theta) \ddot{\theta} + v(\theta, \dot{\theta}) + c_2(\theta)$$

$$M(\theta) = \begin{bmatrix} m_1 l_1^2 + m_2 (l_1 + l_2 c_2)^2 & 0 \\ 0 & m_2 l_2^2 \end{bmatrix}$$

$$v(\theta, \dot{\theta}) = \begin{bmatrix} -2(l_1 + l_2 c_2) m_2 l_2 s_2 \dot{\theta}_1 \dot{\theta}_2 \\ (l_1 + l_2 c_2) m_2 l_2 s_2 \dot{\theta}_1^2 \end{bmatrix}$$

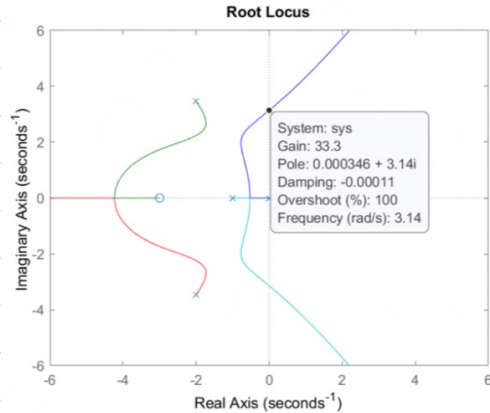
$$G(\theta) = \begin{bmatrix} 0 \\ m_2 g l_2 c_2 \end{bmatrix}$$

Assuming $l_1 = l_2 = m_1 = m_2 = 1$



7) limiting value for K before the loop becomes unstable is 33.3 as seen from the graph below.

```
num = [0 0 0 1 3];
den = [1 5 20 16 0];
rlocus(num, den)
v = [-6 6 -6 6]
axis(v)
```



8) Given,

$$G(s) = 1/(s-1)^2$$

$$C(s) = \frac{K(s+2)}{(s+p)} \quad \left. \vphantom{\frac{K(s+2)}{(s+p)}} \right\} \text{compensator ; } p > 2$$

$$\text{final transfer function } F(s) = \frac{G \cdot C}{1 + G \cdot C}$$

$$F(s) = \frac{K(s+2)}{s^3 + (p-2)s^2 + (K-2p+1)s + K+P}$$

Manually choosing poles at $-4, -5, -6$.

$$\text{Then den. is } s^3 + 15s^2 + 74s + 120$$

$$\text{So, } p-2 = 15 ; K-2p+1 = 74 ; K+P = 120$$

By divided $F(s)$ by dc gain $F(s)$ to stabilize the response at 1. The final step response of $F(s)$ and step response of $G(s)$ are shown below.

$$P = 17, K = 107, Z = 0.9626.$$

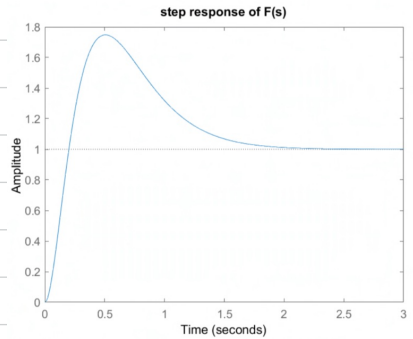
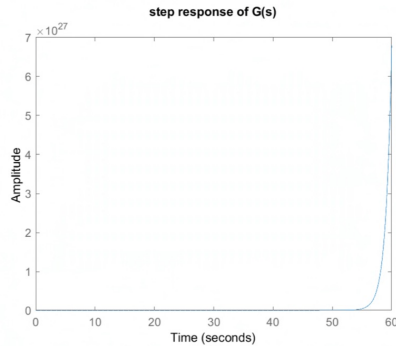
```

s = tf('s');
k = 107;
z = 0.9626;
p = 17;

lead = k * (s + z) / (s + p);
F = 1 / (s - 1)^2;

final_f = feedback(lead * F, 1);
step(final_f / dcgain(final_f))

```



In state space representation,

$$G(s) = \frac{y(s)}{u(s)} = \frac{1}{(s-1)^2} = \frac{1}{s^2 - 2s + 1}$$

$$I^{-1} [y(s) [s^2 - 2s + 1]] = I^{-1} [u(s)]$$

$$\ddot{y} - 2\dot{y} + y = u$$

Now, let $y = x_1$; $\dot{y} = \dot{x}_1 = x_2$; $\ddot{y} = 2x_2 - x_1 + u$

$$\dot{x} = \begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -1 & 2 \end{bmatrix} x + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u ; \quad y = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} x + \begin{bmatrix} 0 \\ 0 \end{bmatrix} u$$

Now,

$$A = \begin{bmatrix} 0 & 1 \\ -1 & 2 \end{bmatrix} ; B = \begin{bmatrix} 0 \\ 1 \end{bmatrix} ; C = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} ; P = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

then, $u = r + Kx$ (Pole placement method)

$$\text{So, } \dot{x} = (A+BK)x + Br ; K = [K_1 \ K_2]$$

$$A_{CL} = A+BK = \begin{bmatrix} 0 & 1 \\ -1+K_1 & 2+K_2 \end{bmatrix}$$

then, finding eigen values of A_{CL}

$$\det(A_{CL} - \lambda I) = 0$$

$$\begin{bmatrix} -\lambda & 1 \\ -1+K_1 & 2+K_2-\lambda \end{bmatrix} = 0$$

$$\lambda^2 - (K_2+2)\lambda + (1-K_1) = 0$$

Manually placing poles at $-4, -5$

$$(\lambda+4)(\lambda+5) = \lambda^2 + 9\lambda + 20$$

$$-(K_2+2) = 9 \rightarrow K_2 = -11$$

$$1 - K_1 = 20 \rightarrow K_1 = -19$$

$$\text{So, } A_{CL} = A + BK = \begin{bmatrix} 0 & 1 \\ -20 & -9 \end{bmatrix}$$

$$\text{where, } r = 1 / \text{dcgain}(\text{system}) = 0.05$$

Step responses before and after stabilization are shown below.

```
A = [0 1; -1 2];
B = [0; 1];
C = [1 0; 0 0];
D = [0; 0];
sys = ss(A, B, C, D);

K = place(A, B, [-4, -5]);
Ac1 = A - B * K;
sys_cl = ss(Ac1, B, C, D);

r = 1 / dcgain(sys_cl);
sys_cl = ss(Ac1, B*r(1), C, D);
step(sys_cl)
```

